

ABIVARSAN KETHEESWARAN

Software Engineer

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Personal profile

Experience in full-stack development, gained through coursework and a significant software project, using agile methodologies. Motivated to learn and contribute, seeking a software Engineer position to enhance technical expertise, collaborate with professionals, and contribute to impactful software solutions.

Education

University of Moratuwa

BSc (Hons) Information Technology and Management

March 2022 – December 2026

Moratuwa, Sri Lanka

St. John's College

G.C.E. A/L – Physical Science Stream

January 2011 – August 2019

Jaffna, Sri Lanka

Technical Skills

Programming Languages: C, C#, Java

Web Technologies: HTML, CSS, JavaScript, TypeScript, Material UI

Frameworks: ASP.NET Core, Spring Boot, React

Databases: MS SQL, MySQL

Cloud Services: Firebase

Version Control: Git

Tools: VS Code, Visual Studio, IntelliJ, GitHub, Postman, Figma, Azure DevOps

Work Experience

Minox Techno

Software Engineer Intern

Mar 2025 – Aug 2025

Jaffna, Sri Lanka

- Gained hands-on experience in semiconductor software systems supporting foundry, memory, and advanced packaging environments.
- Contributed to projects including **CAPCON**, **ASMPYSIM**, and **BSEI** focusing on manual testing, software testing, and host application development.
- Validated system behavior, identified defects, and improved software reliability through structured testing workflows.
- Collaborated with cross-functional R&D teams using task-based development processes.

Projects

Smartwatch Calorie Burn Forecasting & Nutrition Allocation — AI/ML Research | [GitHub](#) 📄

2026

- Built a **Conv1D-LSTM + XGBoost** hybrid model for next-day calorie burn forecasting from smartwatch data, achieving $R^2 = 0.9963$ and **MAE = 27.08 kcal** using clinically validated TDEE targets derived from Mifflin-St Jeor, Keytel, and ACSM peer-reviewed equations.
- Designed a **Dynamic Inverse-Variance MSE** ensemble blending strategy with **SHAP XAI** explainability and an automated BMI-driven macronutrient allocation engine, delivering personalized daily Carbohydrate, Protein, and Fat recommendations from raw wearable sensor data.
- **Technologies:** Python, TensorFlow/Keras, XGBoost, Scikit-learn, SHAP, Pandas, NumPy

Robot Configuration Generator — MVVM | [GitHub](#) 📄

2025

- Designed and developed a desktop application using **WPF** and the **MVVM** architectural pattern.
- Implemented software-hardware integration using simulators to configure and validate robot parameters.
- Improved understanding of industrial automation workflows and system-level application design.
- **Technologies:** WPF, MVVM, .NET

Workforce Management System — 3rd Year Software Project | [GitHub](#) 📄

2025

- Developing a comprehensive workforce platform with task management, time tracking, performance analytics, and real-time updates.
- Inspired by tools such as Asana, Trello, and Slack to improve team productivity and collaboration.
- **Technologies:** React (TypeScript), ASP.NET Core, MS SQL, Firebase

Uni Tutor Platform — 2nd Year Software Project | [Front-end](#) 📄 | [Back-end](#) 📄

2024

- Collaborated with a team of five to design and develop a platform connecting students with university lecturers and undergraduates.
- Implemented features including user registration, profiles, subject management, chat system, search and filtering, reviews, ratings, coin bank, comments, and to-do sections.

- **Contribution:** Developed an Admin Dashboard to verify tutors via university ID, manage user suspension and restoration, view reports, and analyze platform data using charts.
 - **Technologies:** React (TypeScript), ASP.NET Core, MS SQL, Firebase
- TechHub E-Commerce Platform — Microservices Architecture** | *Front-end*  | *Back-end*  **2024**
- Collaborated in a team of six to develop an enterprise-level e-commerce system using microservices architecture.
 - Designed and implemented the Order Service, including order placement, payment processing, order completion, and tracking.
 - Configured Kafka for notification services, integrated Stripe for payments, set up API Gateway, and tested services using Postman.
 - **Technologies:** React.js, Spring Boot, MySQL, Firebase, Docker, Kafka, Zookeeper
- Employee Management Application — MVC** | *GitHub*  **2024**
- Developed an ASP.NET Core MVC application enabling administrators to add, edit, and remove employees.
 - Implemented a working-days calculation feature excluding holidays.
 - **Technologies:** ASP.NET Core MVC
- Top Speed Automobile — MVC** | *GitHub*  **2024**
- Developing a vehicle advertisement application allowing users to view vehicle details and admins to manage listings.
 - **Technologies:** ASP.NET Core MVC
- Task Manager Application — MVC** | *GitHub*  **2024**
- Developing a task management system with user authentication, password recovery, and CRUD operations for tasks.
 - **Technologies:** ASP.NET Core MVC
- QA Automation Test Suite for Web Application — Testing** | *GitHub*  **2025**
- Developing automated test suites covering end-to-end user workflows, API endpoints, and regression scenarios using BDD methodology.
 - **Technologies:** Playwright, Cucumber, JavaScript/TypeScript, UI Testing, API Testing, BDD

Soft Skills

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|------------------------|-----------------------|-------------|
| • Verbal Communication | • Attention to Detail | • Team Work |
| • Problem Solving | • Responsiveness | |

Languages

- | | |
|---|-----------------|
| • English(Professional Working Proficiency) | • Tamil(Native) |
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Interests

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| • Playing Football | • Browsing Internet |
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Extracurricular Activities

Secretary, Science Union 2018, St. John’s College, Jaffna
 Secretary, St. John’s Ambulance 2018, St. John’s College, Jaffna
 Industry Visit and Organizing Team Building Event, Organizing Committee 2023, Creative Software Participation, HackMoral 6.0 2024, INTECS, Batch21, FIT, UOM
 Participation, Monthly Code Surge 2024, Pagero
 Participation, AlgoXplore 1.0 2024, NSBM Green University
 Representative, Final Year, ITM, University of Moratuwa

References

Miss. Maathangi Baskaran Senior Software Engineer Mindox Techno, Jaffna, Sri Lanka Email: maathangi.baskaran@mindoxtechno.com Tel: +94 76 747 9308	Dr. Lochandaka Ranathunga, Senior Lecturer Faculty of Information Technology University of Moratuwa Email: lochandaka@uom.lk Tel: +94 11 2650301
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